

# Deriving the Multi-Dose Concentration Formula

from a first-order differential equation  
via an infinite geometric series

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Beginning with the elimination law  $c'(t) = k_1 c(t)$ , we construct the single-dose profile, superpose  $n$  doses to obtain a finite geometric series, and pass to the limit  $n \rightarrow \infty$ . The steady-state peak and trough then follow by evaluating one formula at two points, and the finite- $n$  expression reproduces the MANYDOSECONCENTRATION function exactly.

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## Roadmap

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- Step 1.** Solve the differential equation and reparameterise by half-life
- Step 2.** Build the single-dose profile and the tail of dose  $k$
- Step 3.** Superpose  $n$  doses: a finite geometric series
- Step 4.** Let  $n \rightarrow \infty$ : the series converges
- Step 5.** Evaluate at  $\tau = 0$  and  $\tau = P - x$ : peak and trough
- Step 6.** Recover the MANYDOSECONCENTRATION function and verify against the notebooks
- Step 7.** Take the continuous limit: dosing frequency  $D \rightarrow \infty$

Symbols:  $t$  time (h)  $\cdot$   $h$  elimination half-life (h)  $\cdot$   $x$  absorption duration (h)  $\cdot$   $P$  dosing period (h, default 24)  $\cdot$   $D$  number of doses per  $P$  hours  $\cdot$   $\tau$  time since the current dose finished absorbing.

## Step 1. Solve the Differential Equation

A drug is cleared at a rate proportional to the amount present. This first-order law is the entire physical input to the model:

$$c'(t) = k_1 c(t), \quad k_1 < 0. \quad (1)$$

Separating variables gives the general solution

$$c(t) = k_2 e^{k_1 t}. \quad (2)$$

### Reparameterising by half-life

The constant  $k_1$  is not measured clinically; the *half-life*  $h$  is. Imposing  $c(t+h) = \frac{1}{2} c(t)$  on (2),

$$e^{k_1 h} = \frac{1}{2} \iff k_1 h = -\ln 2 \iff k_1 = -\frac{\ln 2}{h}. \quad (3)$$

Substituting (3) into (2) and using  $e^{-(\ln 2)t/h} = 2^{-t/h}$  converts the solution to base two:

$$c(t) = k_2 \cdot 2^{-t/h} \quad (4)$$

#### Why base two

In the form  $2^{-t/h}$  the half-life is read off directly: one half-life ( $t = h$ ) gives  $2^{-1} = \frac{1}{2}$ , two give  $2^{-2} = \frac{1}{4}$ , and  $m$  give  $2^{-m}$ . Every subsequent formula inherits this base, which is why powers of 2 rather than  $e$  appear throughout.

## Step 2. The Single-Dose Profile

An oral dose is not absorbed instantaneously. Over an absorption window of length  $x$  the drug enters the blood at a constant rate; afterwards (4) governs its decay. Normalising so that one dose reaches unit concentration at  $t = x$ :

$$c_{\text{single}}(t) = \begin{cases} \frac{t}{x}, & 0 < t \leq x \quad (\text{absorption}), \\ 2^{-(t-x)/h}, & t > x \quad (\text{elimination}). \end{cases} \quad (5)$$

Both boundary conditions hold by construction:  $c_{\text{single}}(0) = 0$  and  $c_{\text{single}}(x) = 1$ .

### The tail left by dose $k$

Doses are given every  $P$  hours (or more generally,  $D$  doses every  $P$  hours, so the interval is  $P/D$ ). Dose  $k$  is administered at time  $(k-1)P$  (for  $D=1$ ) and finishes absorbing at

$$t_k = (k-1)P + x. \quad (6)$$

For any  $t > t_k$  its residual contribution is the elimination branch of (5) translated to start at  $t_k$ :

$$c_k(t) = 2^{-(t-t_k)/h} = 2^{-(t-x-(k-1)P)/h}. \quad (7)$$

Each past dose therefore leaves its own exponential tail, and successive tails are offset from one another by exactly one period  $P$ .

### Step 3. Superposing $n$ Doses

Equation (1) is linear, so contributions add. After  $n$  doses, at a time  $t$  lying in the elimination phase of the current cycle,

$$C_n(t) = \sum_{k=1}^n c_k(t) = \sum_{k=1}^n 2^{-(t-x-(k-1)P)/h}. \quad (8)$$

This sum can be factored in two equivalent ways. Both are useful, and the distinction between them matters.

#### Form A — anchored at the first dose

Splitting the exponent in (8) and setting  $j = k - 1$ ,

$$C_n(t) = 2^{-(t-x)/h} \sum_{j=0}^{n-1} (2^{P/h})^j = 2^{-(t-x)/h} \cdot \frac{2^{nP/h} - 1}{2^{P/h} - 1}. \quad (9)$$

Here the ratio is  $2^{P/h} > 1$ : measured from the *first* dose, later terms are larger. Form A is the expression implemented in the notebooks.

#### Form B — anchored at the most recent dose

Let  $\tau$  denote elapsed time since the *current* dose finished absorbing,

$$\tau = t - t_n = t - x - (n - 1)P, \quad 0 \leq \tau \leq P - x. \quad (10)$$

Re-indexing (8) by  $i = n - k$  (cycles into the past) gives

$$C_n(\tau) = \sum_{i=0}^{n-1} 2^{-(\tau+iP)/h} = 2^{-\tau/h} \sum_{i=0}^{n-1} r^i, \quad r := 2^{-P/h} \in (0, 1). \quad (11)$$

Now the ratio is  $r < 1$ , and the standard partial sum applies:

$$\sum_{i=0}^{n-1} r^i = \frac{1 - r^n}{1 - r} \quad \Rightarrow \quad C_n(\tau) = 2^{-\tau/h} \cdot \frac{1 - 2^{-nP/h}}{1 - 2^{-P/h}} \quad (12)$$

#### The two forms are the same

Substituting  $t = (n - 1)P + x + \tau$  into (9) and multiplying through by  $2^{P/h}/2^{P/h}$  recovers (12) identically:

$$2^{-(t-x)/h} \frac{2^{nP/h} - 1}{2^{P/h} - 1} = 2^{-\tau/h} \frac{2^{P/h} - 2^{-(n-1)P/h}}{2^{P/h} - 1} = 2^{-\tau/h} \frac{1 - 2^{-nP/h}}{1 - 2^{-P/h}}.$$

Only Form B has a finite limit as  $n \rightarrow \infty$ , because it holds the observation point  $\tau$  fixed *within* the current cycle rather than letting  $t$  run away with  $n$ . Mixing the prefactor of one form with the ratio of the other is the one algebraic trap in this derivation.

#### Step 4. The Limit $n \rightarrow \infty$

Since  $0 < r < 1$  we have  $r^n = 2^{-nP/h} \rightarrow 0$ , so (12) converges:

$$C_\infty(\tau) = \lim_{n \rightarrow \infty} C_n(\tau) = \frac{2^{-\tau/h}}{1 - 2^{-P/h}} = \frac{2^{(P-\tau)/h}}{2^{P/h} - 1} \quad (13)$$

Written out, the limit is a plain geometric sum of the tails of all past doses:

$$\begin{aligned} C_\infty(\tau) &= 2^{-\tau/h} + 2^{-(\tau+P)/h} + 2^{-(\tau+2P)/h} + \dots \\ &= 2^{-\tau/h} \left( 1 + 2^{-P/h} + 2^{-2P/h} + \dots \right) \\ &= \frac{2^{-\tau/h}}{1 - 2^{-P/h}}. \end{aligned} \quad (14)$$

#### The accumulation multiplier

The factor  $(1 - 2^{-P/h})^{-1} = 2^{P/h} / (2^{P/h} - 1)$  measures how much steady-state dosing amplifies a single dose.

- $h \gg P$  (slow clearance):  $2^{P/h} \approx 1 + \frac{P \ln 2}{h}$ , so the multiplier is  $\approx \frac{h}{P \ln 2} \gg 1$  — strong accumulation.
- $h \ll P$  (fast clearance):  $2^{P/h} \gg 1$  and the multiplier tends to 1 — each dose has essentially cleared before the next.

#### Step 5. Peak and Trough

Formula (13) decreases monotonically in  $\tau$ , so the extremes of a steady-state cycle occur at the two endpoints of  $\tau \in [0, P - x]$ .

##### Peak: $\tau = 0$

At  $\tau = 0$  the current dose has just reached unit concentration and all earlier tails are at their largest:

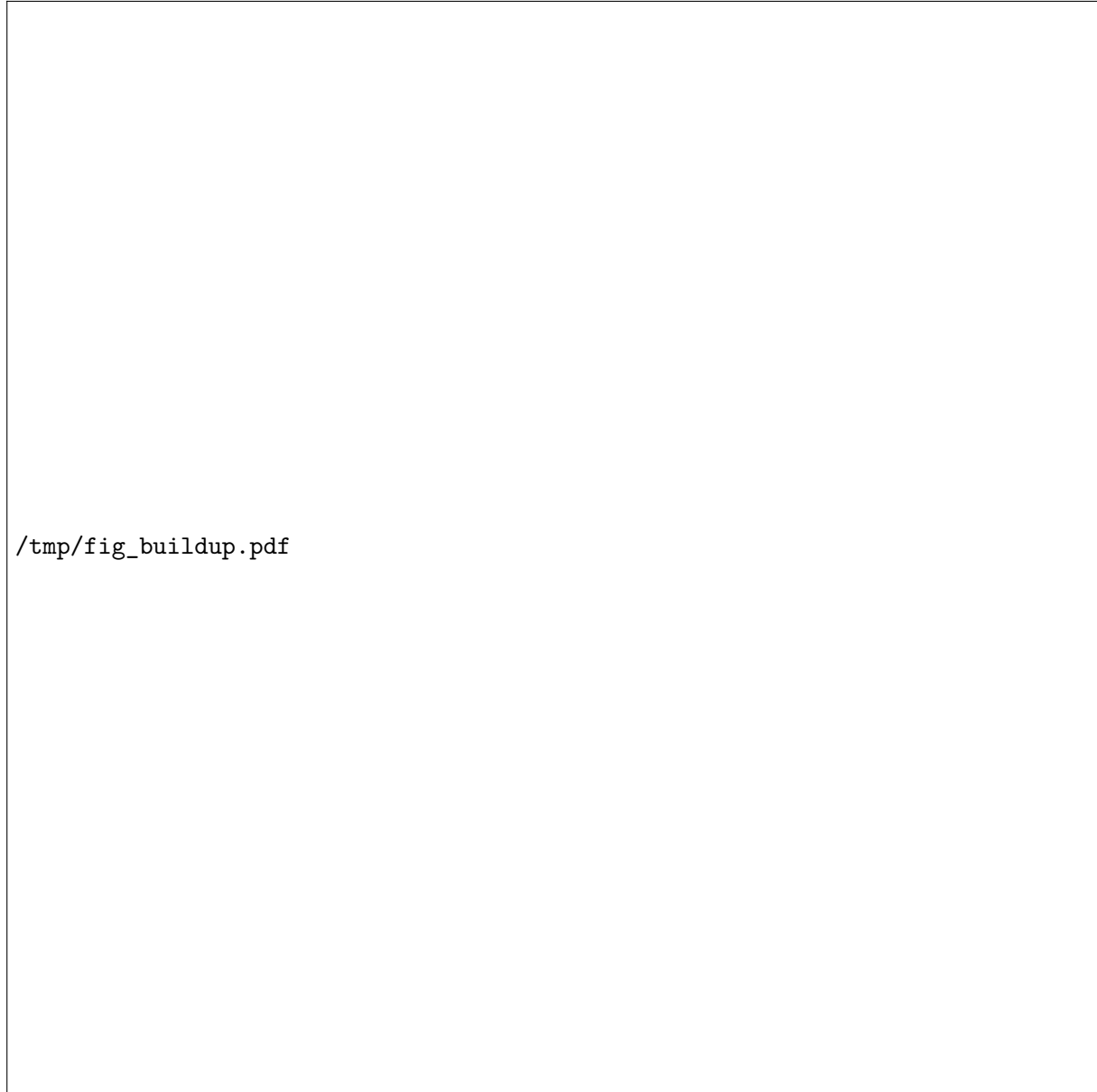
$$\text{peak}_{\text{ss}} = C_\infty(0) = \frac{1}{1 - 2^{-P/h}} = \frac{2^{P/h}}{2^{P/h} - 1} = 1 + \frac{1}{2^{P/h} - 1}. \quad (15)$$

The final form separates the two physical contributions: the 1 is the dose just absorbed, and  $(2^{P/h} - 1)^{-1}$  is everything left over from all previous doses.

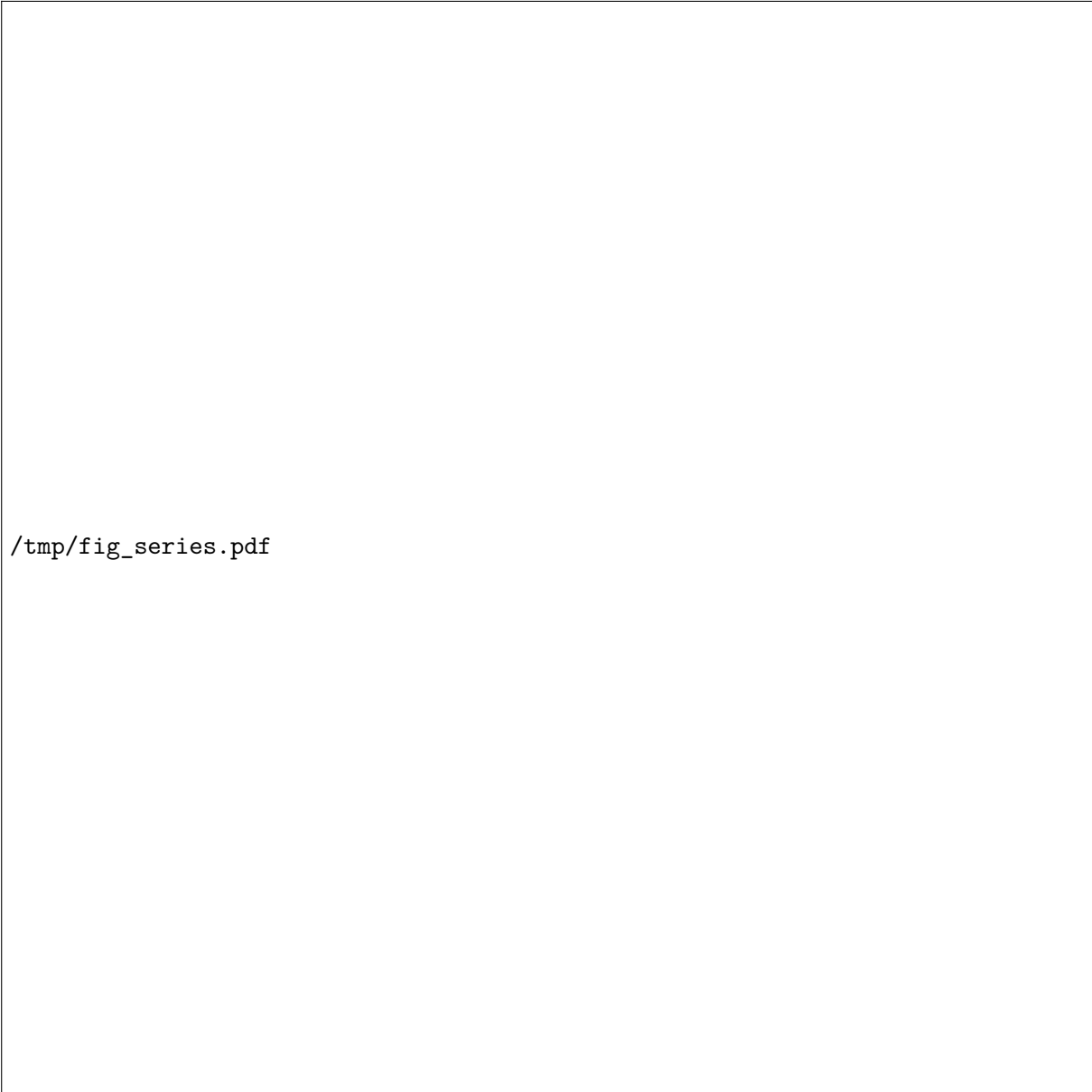
##### Trough: $\tau = P - x$

The minimum falls the instant before the next dose, i.e.  $\tau = P - x$ :

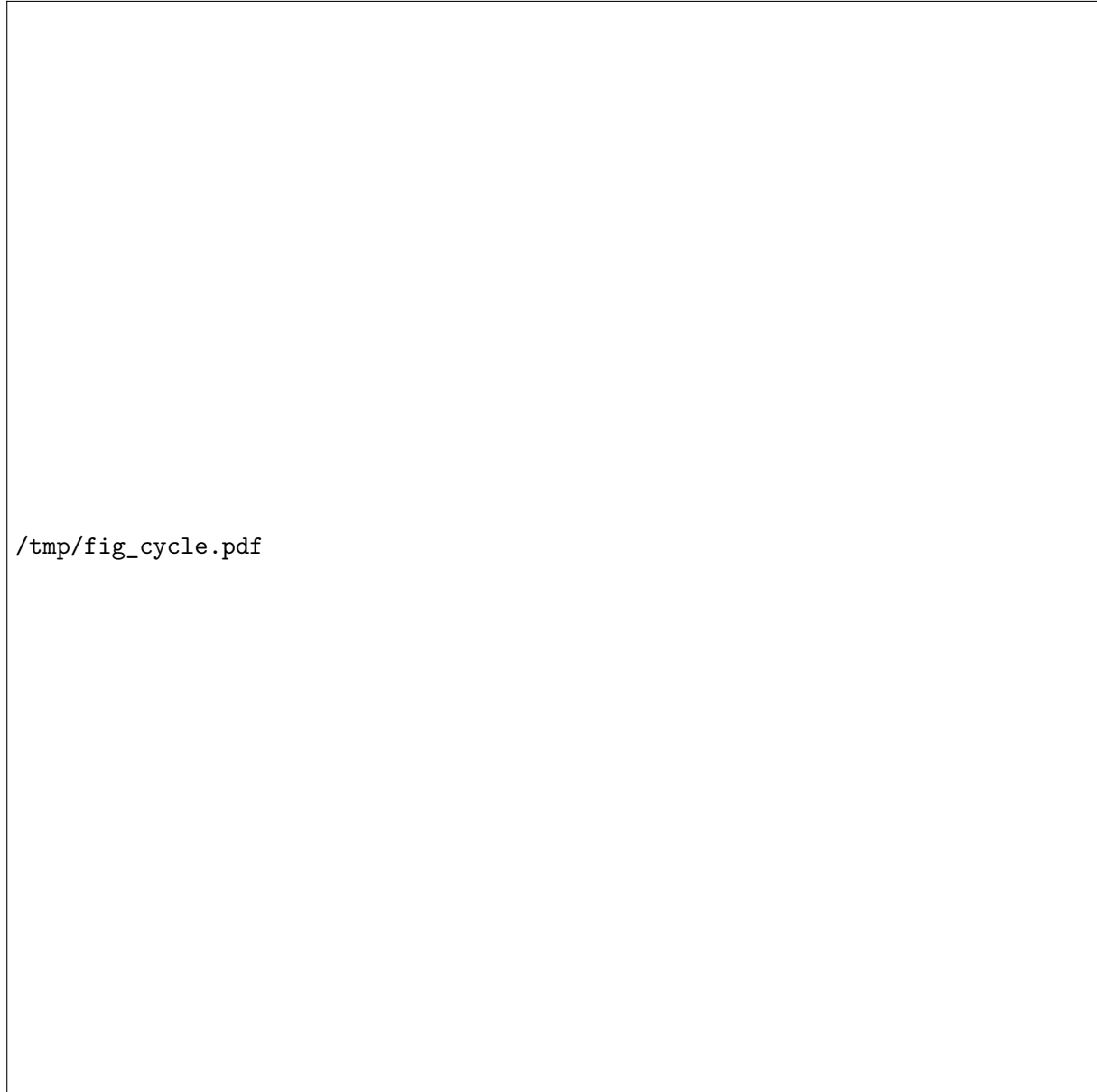
$$\text{trough}_{\text{ss}} = C_\infty(P - x) = \frac{2^{-(P-x)/h}}{1 - 2^{-P/h}} = \frac{2^{x/h}}{2^{P/h} - 1}, \quad (16)$$



**Figure 1:** Successive partial sums  $C_n$  for  $x = 4$  h,  $h = 20$  h,  $P = 24$  h. Each curve adds one dose; the profile converges to a periodic steady state bounded by the horizontal lines derived in Step 5.



**Figure 2:** Left: the contributions entering the trough form a geometric sequence with ratio  $r = 2^{-P/h}$ . Right: their partial sums converge to the closed form  $2^{x/h} / (2^{P/h} - 1)$ .



**Figure 3:** A single steady-state cycle. The maximum occurs the instant absorption ends ( $\tau = 0$ ) and the minimum the instant before the next dose ( $\tau = P - x$ ).



the last step following on multiplying numerator and denominator by  $2^{P/h}$ . No approximation is involved: the  $2^{x/h}$  arises purely from the offset between the end of absorption and the next dose.

### Fluctuation index

Dividing (15) by (16), the shared denominator cancels and the ratio collapses to a single exponential:

$$\frac{\text{peak}_{\text{ss}}}{\text{trough}_{\text{ss}}} = \frac{2^{P/h}}{2^{x/h}} = 2^{(P-x)/h} \quad (17)$$

The swing between peak and trough depends only on how many half-lives fit into the gap  $P - x$  between the end of absorption and the next dose. Extending the release window (larger  $x$ ) or shortening the interval (smaller  $P$ ) flattens the profile; a short half-life sharpens it.

#### Steady-state results

$$\text{peak}_{\text{ss}} = \frac{2^{P/h}}{2^{P/h} - 1} = 1 + \frac{1}{2^{P/h} - 1}, \quad \text{trough}_{\text{ss}} = \frac{2^{x/h}}{2^{P/h} - 1}, \quad \frac{\text{peak}_{\text{ss}}}{\text{trough}_{\text{ss}}} = 2^{(P-x)/h}.$$

### Agreement with the notebook definitions

The notebooks define, with  $P = 24$ ,

$$\text{troughmin}[x, h] = \frac{2^{x/h}}{2^{24/h} - 1}, \quad (18)$$

$$\text{peakmax}[x, h] = \frac{1}{2^{24/h} - 1} + \frac{(12 + x - 12 \lceil x/24 \rceil) \lceil x/24 \rceil}{x}. \quad (19)$$

Equation (18) is exactly (16). For (19), note that whenever  $0 < x \leq 24$  we have  $\lceil x/24 \rceil = 1$ , so the second term reduces to

$$\frac{(12 + x - 12) \cdot 1}{x} = \frac{x}{x} = 1,$$

and (19) becomes  $1 + (2^{24/h} - 1)^{-1}$ , precisely (15). The ceiling expression is therefore not an extra physical effect but the generalisation that keeps the formula valid when the absorption window spans more than one dosing period,  $x > P$ .

## Step 6. Recovering the ManyDoseConcentration Function

For finitely many doses the notebooks evaluate Form A, (9), using ceilings to identify the current cycle. With  $P = 24$  set

$$n = \lceil t/24 \rceil, \quad m = \lceil (t - x)/24 \rceil, \quad (20)$$

so that  $n$  counts doses administered and  $m$  counts those that have finished absorbing. The complete profile is

$$C(t; x, h) = \begin{cases} \frac{(12 + t - 12n)n}{x}, & 0 < t \leq x, \\ \underbrace{2^{(x-t)/h} \frac{2^{24m/h} - 1}{2^{24/h} - 1}}_{\text{Form A, eq. (9)}} - \underbrace{\frac{(n-m)(-12-t+12n+12m)}{x}}_{\text{absorption-ramp correction}}, & t > x. \end{cases} \quad (21)$$

#### Where the series sits in the code

The first bracket of the lower branch is (9) verbatim, with  $nP$  replaced by  $24m$ : it is the finite geometric series summing the tails of the  $m$  doses that have completed absorption. Letting  $m \rightarrow \infty$  with  $\tau$  fixed returns (13), the steady state. The second bracket subtracts the portion of the linear absorption ramps already accounted for, and vanishes whenever  $n = m$  — that is, whenever no dose is mid-absorption.

### Step 7. Continuous IV Dosing

To extend to more general dosing schedules, consider  $D$  doses given every  $P$  hours. The inter-dose interval is  $\Delta = P/D$ . As  $D \rightarrow \infty$  (infinite dosing frequency), we approach continuous intravenous infusion at rate  $\lambda = D/P$  (doses per hour).

All previous formulas remain valid with  $P$  interpreted as the dosing period and  $D$  as the number of doses per period. In the continuous limit, replace  $P$  with  $P/D$  (the inter-dose interval) in the exponents:  $2^{P/h}$  becomes  $2^{P/(Dh)}$ .

For instantaneous absorption ( $x \rightarrow 0$ ) and continuous dosing ( $D \rightarrow \infty$ ), the steady-state concentration approaches

$$C_{\infty, \text{cont}} = \frac{\lambda h}{\ln 2}, \quad (22)$$

where  $\lambda = D/P$  is the continuous dosing rate. This recovers the classical pharmacokinetic result:  $C = R/k$  with  $R = \lambda$  (infusion rate) and  $k = \ln 2/h$  (elimination rate constant).

### Summary

| Step | Result                                             | Mathematics                                                              |
|------|----------------------------------------------------|--------------------------------------------------------------------------|
| 1    | Exponential solution                               | $c' = k_1 c \implies c(t) = k_2 2^{-t/h}$                                |
| 2    | Single-dose profile                                | ramp $t/x$ on $(0, x]$ , then $2^{-(t-x)/h}$                             |
| 3    | Superposition of $n$ doses                         | $C_n(\tau) = 2^{-\tau/h} (1 - r^n)/(1 - r)$ , $r = 2^{-P/h}$             |
| 4    | Convergence as $n \rightarrow \infty$              | $C_\infty(\tau) = 2^{-\tau/h} / (1 - 2^{-P/h})$                          |
| 5    | Peak, trough, fluctuation                          | $2^{P/h} / (2^{P/h} - 1)$ , $2^{x/h} / (2^{P/h} - 1)$ ,<br>$2^{(P-x)/h}$ |
| 6    | Finite- $n$ form is MANY-DOSECONCENTRATION         | ceilings $n = \lceil t/24 \rceil$ , $m = \lceil (t-x)/24 \rceil$         |
| 7    | Continuous IV dosing<br>( $D \rightarrow \infty$ ) | $C_{\infty, \text{cont}} = \lambda h / \ln 2$                            |

The derivation reduces to one observation: a sum of exponential tails offset by equal time steps *is* a geometric series. Once the sum is anchored at the most recent dose — so that the observation point stays inside the current cycle — the closed form, its limit, and both clinical extremes follow in a few lines of algebra.

Source: *drugs from scratch* Mathematica notebooks, Levels 1–8 (including the Cymbalta application), and the Python translation `ManyDoseConcentrationdrugs2.ipynb`.